GeoODE-KD: Distilling Sentence Embeddings as Teacher-Guided Geometric Dynamics

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Abstract

Embedding distillation can become restrictive when intermediate supervision treats representations at different depths as independent matches to the same final teacher embedding. Such objectives specify target states, but leave the transitions between them undefined. We propose **GeoODE-KD**, a sentence embedding distillation framework that turns cached teacher outputs into depth-dependent transition targets. GeoODE-KD places normalized embeddings on a hypersphere and defines a teacher-conditioned potential from point-wise semantic alignment and native teacher geometry. Its negative Riemannian gradient gives the direction of semantic change, while a finite-horizon schedule determines how much of the remaining displacement each student layer should cover. GeoODE-KD does not insert a Neural ODE module or run a numerical solver; instead, it treats the existing Transformer layers as a discrete trajectory and trains every intermediate transition to move along the teacher-conditioned vector field. The method uses only final teacher embeddings cached before training and leaves the inference architecture unchanged. Its ideal continuous flow preserves embedding norms and decreases the teacher-conditioned energy; without the relational term, geodesic distance to the teacher contracts in closed form and reaches zero at the final depth.

1 Introduction

Dense text embeddings support semantic search, retrieval-augmented generation, clustering, classification, and semantic textual similarity. The strongest embedding models, however, are often too large or expensive for routine deployment. Knowledge distillation addresses this cost by training a compact student from a stronger teacher (Hinton et al., 2015; Reimers & Gurevych, 2020). Most embedding distillation pipelines still treat this transfer as a state-matching problem: given a teacher embedding, make one or more student representations resemble it.

Prior work improves state matching in several ways, but each leaves the path through student depth unspecified. Output-level methods align the final embedding and add contrastive supervision (Xu et al., 2023; Zhang et al., 2024). Relational and optimal-transport objectives also transfer pairwise or token-level structure (Truong et al., 2025). These methods provide richer terminal supervision, but they do not determine how intermediate sentence representations should change. Depth-aware methods supervise several student layers or propagate structure between adjacent layers (Li et al., 2025; Dao et al., 2026). They expose more of the student to the teacher signal, but selected layers are still trained as separate target states.

This formulation becomes restrictive when teacher and student have different capacities. A shallow student layer must retain information needed by later computation, while a final teacher embedding already represents the end of a much deeper transformation. Pulling many shallow layers toward that same endpoint can flatten the student's internal hierarchy: the model is asked to produce terminal semantics before it has completed the computation needed to support them. Choosing fewer anchor layers reduces the constraint, but still does not specify what should happen between two anchors. The missing signal is not another target state; it is a rule for moving from the current state to the next one.

We propose **GeoODE-KD**, a sentence embedding distillation framework that makes this transition rule explicit. GeoODE-KD views normalized layer representations as points on a hypersphere and uses cached teacher embeddings to define a semantic potential. A point-wise term attracts each sentence to its teacher target, while a relational term preserves the teacher's native cosine geometry. The negative Riemannian gradient of this potential gives a local direction of semantic improvement. A finite-horizon schedule then assigns each student layer a fraction of the displacement that remains before the output.

GeoODE-KD does not replace Transformer blocks with a Neural ODE or train a separate velocity network. Instead, it reads the existing Transformer layers as a discrete trajectory. At depth t_ℓ, the teacher-conditioned field gives the direction in which the state should move toward $t_{\ell+1}$. The next Transformer layer is trained so that its realized update points along this field. Intermediate layers are therefore supervised by transitions, $\mathbf{Z}^{(\ell)} \to \mathbf{Z}^{(\ell+1)}$, rather than independent copies of the terminal condition, $\mathbf{Z}^{(\ell)} \to \mathbf{T}$. This use of continuous dynamics is a training principle, not a change to the student architecture (Chen et al., 2018; Li et al., 2022).

The construction also separates teacher cost from student optimization. The teacher is run once, and only its final sentence embeddings are cached. GeoODE-KD requires no teacher hidden states, attention maps, learned projection head, or numerical ODE solve during training. The geometric machinery is discarded after training, leaving the original student encoder at inference. The formulation further makes the claimed behavior measurable across depth: the ideal flow preserves embedding norms and decreases teacher-conditioned energy, while its instance-only form follows a closed geodesic contraction profile that reaches the teacher at the final depth.

Our contributions are threefold:

- We formulate embedding distillation as trajectory supervision, where the teacher specifies transitions between consecutive student layers rather than a collection of independent target states.
- We derive a finite-horizon Riemannian field that combines point-wise alignment with native teacher geometry and produces an explicit local target at every student depth.
- We train the existing Transformer layers to follow these targets with a velocity-matching objective that uses only cached teacher outputs and adds no inference-time parameters.

2 Background and Related Work

2.1 Static Embedding Distillation

Let $\mathbf{z}_i^{(\ell)}$ be the normalized representation of sentence x_i after student layer ℓ, and let $\boldsymbol{\tau}_i$ be its normalized teacher target. Endpoint distillation constrains only

$$\mathbf{z}_i^{(L)} \longrightarrow \boldsymbol{\tau}_i. \tag{1}$$

This objective is efficient, but leaves the internal path unconstrained. Multi-layer distillation adds the same target at selected depths $\mathcal{A}$,

$$\mathcal{L}_{\text{layer}} = \sum_{\ell \in \mathcal{A}} D\Big(\mathbf{Z}^{(\ell)}, \mathbf{T}\Big)\,, \tag{2}$$

where $\mathbf{Z}^{(\ell)}$ and $\mathbf{T}$ stack a mini-batch of student and teacher embeddings. Relational variants add discrepancies between pairwise similarity matrices. All three cases supervise states: endpoint KD constrains one state, multi-layer KD constrains several states, and relational KD constrains their geometry.

Equation 2 does not say how much semantic change is appropriate at layer ℓ. Weak supervision may have little effect before the output; strong supervision may force shallow features to imitate a state that the teacher produces only after substantially more computation. Selecting fewer anchors reduces this tension but still leaves the transitions undefined.

2.2 Depth as Semantic Time

We index student representations by normalized depth,

$$t_\ell = \frac{\ell}{L}, \qquad \mathbf{Z}^{(\ell)} \approx \mathbf{Z}(t_\ell), \tag{3}$$

and treat consecutive layers as observations along a semantic trajectory. A teacher-conditioned vector field then specifies the local transition,

$$\frac{d\mathbf{Z}(t)}{dt} = F(\mathbf{Z}(t), \mathbf{T}, t)\,. \tag{4}$$

Static matching asks whether $\mathbf{Z}^{(\ell)}$ resembles the terminal teacher state. Equation 4 instead asks whether the transition $\mathbf{Z}^{(\ell)} \rightarrow \mathbf{Z}^{(\ell+1)}$ moves in a direction and at a pace that reduce teacher discrepancy.

Semantic time is an ordered coordinate for supervision; it is not an assumption that a pretrained Transformer is literally the solution of a latent ODE. The student remains a conventional finite-depth encoder. Continuous dynamics only provide compatible local targets whose composition leads toward the endpoint.

2.3 Related Distillation Paradigms

Sentence-embedding distillation. Direct regression is attractive because teacher sentence embeddings can be cached and reused (Reimers & Gurevych, 2020). Contrastive and multi-stage methods enrich this signal (Xu et al., 2023; Zhang et al., 2024); hidden-state relations and optimal transport address architectural or tokenizer mismatch (Truong et al., 2025). These methods strengthen what is matched. GeoODE-KD instead changes how the target is used: each cached output defines both an endpoint and a direction of evolution.

Layer-wise and geometric transfer. ESE trains depth-scalable representations, while TALAS combines upper-layer teacher anchors with adjacent-layer structural propagation (Li et al., 2025; Dao et al., 2026). Both make depth explicit, but supervise states or relations at selected locations. GeoODE-KD derives each local transition from one teacher-conditioned potential, so relational knowledge changes the direction of motion rather than acting only as another state constraint.

Continuous-depth and flow-based transfer. Neural ODEs replace discrete blocks with a learned differential equation, while ODE-inspired Transformers use integration rules in architecture design (Chen et al., 2018; Li et al., 2022). Diffusion and flow matching also transport student features or distributions toward a teacher (Huang et al., 2023; Lipman et al., 2023; Chen & Lipman, 2024; Shao et al., 2024). GeoODE-KD learns no velocity network and performs no continuous solve. Its field is the analytic gradient of a teacher-conditioned potential, and the existing Transformer layers follow a geometric discretization of that field. The continuous formulation therefore disappears at inference time.

3 GeoODE-KD: From Teacher Embeddings to Semantic Dynamics

Given an unlabeled corpus $\mathcal{D} = \{x_i\}_{i=1}^{N}$, a teacher f_T, and a student f_S with L Transformer layers, our goal is to train the student without labels or online teacher inference. GeoODE-KD follows four steps. It first caches and maps teacher embeddings into the student space. It then converts teacher discrepancy into a tangent vector field, assigns this field a finite depth horizon, and uses the resulting local updates to supervise consecutive student layers. All four steps define training targets; none adds a module to the student.

3.1 Teacher Target Construction

Let

$$f_T : \mathcal{X} \rightarrow \mathbb{R}^{d_T}, \qquad f_S : \mathcal{X} \rightarrow \mathbb{R}^{d_S} \tag{5}$$

denote the teacher and student embedding functions. The offline stage runs f_T once and caches

$$\widetilde{\boldsymbol{\tau}}_i = \mathrm{norm}(f_T(x_i))\,, \qquad \mathrm{norm}(\mathbf{x}) = \frac{\mathbf{x}}{\|\mathbf{x}\|_2 + \epsilon}. \tag{6}$$

No teacher hidden state or attention map is retained.

Point-wise supervision requires both models to use the same width. For $d_T \geq d_S$, we fit a fixed PCA basis $P_{\mathrm{PCA}} \in \mathbb{R}^{d_T \times d_S}$ on the cached training embeddings. This basis selects a maximum-variance d_S-dimensional teacher subspace, but its orientation is arbitrary. The orientation leaves cosine geometry unchanged, yet a point-wise loss would still force the pretrained student to match it.

We fix this coordinate *gauge* before training. Let $\mathbf{T}_{\mathrm{PCA}}$ contain normalized PCA coordinates for a fitting subset, and let $\mathbf{Z}_0^{(L)}$ contain normalized output embeddings of the initial student for the same sentences. We solve orthogonal Procrustes (Schönemann, 1966):

$$R^\star = \underset{R \in O(d_S)}{\arg\min} \left\| \mathbf{T}_{\mathrm{PCA}} R - \mathbf{Z}_0^{(L)} \right\|_F, \qquad R^\star = UV^\top, \quad U\Sigma V^\top = \mathrm{svd}\Big(\mathbf{T}_{\mathrm{PCA}}^\top \mathbf{Z}_0^{(L)}\Big). \tag{7}$$

The shared-space point-wise target is then

$$\boldsymbol{\tau}_i = \mathrm{norm}(f_T(x_i) P_{\mathrm{PCA}} R^\star) \in \mathbb{S}^{d_S - 1}. \tag{8}$$

When $d_T = d_S$, P_{PCA} can be the identity. The rotation $R^\star$ changes neither pairwise cosines nor retained variance. We freeze both maps after fitting; Appendix B gives the target construction details.

For the student, let $\mathbf{h}_i^{(\ell)}$ be the token states after Transformer layer ℓ. Pooling and normalization produce

$$\mathbf{z}_i^{(\ell)} = \mathrm{norm}\Big(\mathrm{Pool}(\mathbf{h}_i^{(\ell)})\Big) \in \mathbb{S}^{d_S - 1}. \tag{9}$$

For a mini-batch of B sentences, we stack these row-wise into $\mathbf{Z}^{(\ell)}, \mathbf{T} \in \mathbb{R}^{B \times d_S}$.

Point-wise alignment uses the projected targets $\mathbf{T}$, but pairwise cosines do not require equal widths. For each mini-batch, we construct the native teacher Gram matrix

$$\mathbf{G}_T = \widetilde{\mathbf{T}} \widetilde{\mathbf{T}}^\top, \tag{10}$$

where $\widetilde{\mathbf{T}}$ stacks the embeddings in Eq. 6. This matrix is independent of teacher width and invariant to $R^\star$, so projection affects only the point-wise target.

3.2 Teacher-Conditioned Vector Field

The next stage turns teacher discrepancy into a direction of motion. For a student state $\mathbf{z}_i$ and projected target $\boldsymbol{\tau}_i$, the hyperspherical distance is

$$d_g(\mathbf{z}_i, \boldsymbol{\tau}_i) = \arccos(\mathbf{z}_i^\top \boldsymbol{\tau}_i). \tag{11}$$

The point-wise potential averages its squared value,

$$\mathcal{E}_{\mathrm{sem}}(\mathbf{Z}, \mathbf{T}) = \frac{1}{2B} \sum_{i=1}^{B} d_g^2(\mathbf{z}_i, \boldsymbol{\tau}_i). \tag{12}$$

Its negative Riemannian gradient points along the shortest geodesic to the teacher target.

The relational potential compares the student Gram matrix with the native teacher matrix from Eq. 10:

$$\mathcal{E}_{\mathrm{geo}}(\mathbf{Z}, \mathbf{G}_T) = \frac{1}{B^2} \left\| \mathbf{Z}\mathbf{Z}^\top - \mathbf{G}_T \right\|_F^2. \tag{13}$$

We combine the two terms as

$$\mathcal{E}(\mathbf{Z}; \mathbf{T}, \mathbf{G}_T) = \alpha \mathcal{E}_{\mathrm{sem}}(\mathbf{Z}, \mathbf{T}) + \beta \mathcal{E}_{\mathrm{geo}}(\mathbf{Z}, \mathbf{G}_T). \tag{14}$$

with $\alpha, \beta \geq 0$. GeoODE-KD does not minimize this expression separately at every layer. It differentiates the potential to obtain the next transition target.

For $\mathbf{z} \in \mathbb{S}^{d_S - 1}$, the projection of an ambient vector $\mathbf{u}$ onto the tangent space is

$$\Pi_{\mathbf{z}}(\mathbf{u}) = \mathbf{u} - (\mathbf{z}^\top \mathbf{u})\mathbf{z}. \tag{15}$$

The hyperspherical logarithmic map is

$$\mathrm{Log}_{\mathbf{z}}(\boldsymbol{\tau}) = \frac{\theta}{\sin\theta} \Pi_{\mathbf{z}}(\boldsymbol{\tau}), \qquad \theta = d_g(\mathbf{z}, \boldsymbol{\tau}). \tag{16}$$

Applying both maps row-wise gives

$$\mathbf{V}(\mathbf{Z}, \mathbf{T}, \mathbf{G}_T) = \alpha \operatorname{Log}_{\mathbf{Z}}(\mathbf{T}) - \frac{4\beta}{B} \Pi_{\mathbf{Z}} \left[(\mathbf{Z}\mathbf{Z}^\top - \mathbf{G}_T)\mathbf{Z} \right]. \tag{17}$$

This direction equals $-B \operatorname{grad}_{\mathbb{S}} \mathcal{E}$. The factor B removes the scale introduced by batch averaging, so flow speed does not change with mini-batch size.

3.3 Finite-Horizon Depth Schedule

A standard gradient flow approaches its target asymptotically, while the student has only L layers. We map these layers to $t \in [0, 1]$. Let $s(t) \geq 0$ allocate guidance over depth, with remaining mass

$$R(t) = \int_t^1 s(u)\, du. \tag{18}$$

The depth-indexed field is

$$\frac{d\mathbf{Z}(t)}{dt} = \frac{s(t)}{R(t)} \mathbf{V}(\mathbf{Z}(t), \mathbf{T}, \mathbf{G}_T), \tag{19}$$

for $t < 1$. The ratio $s(t)/R(t)$ maps unbounded gradient-flow time into this finite interval. We use $s(t) = t$, which gives $R(t) = (1 - t^2)/2$ and places more guidance near the output. Constant and power schedules are direct alternatives.

The instance-only case shows what this schedule controls. For $\alpha = 1$ and $\beta = 0$, let $\theta(t) = d_g(\mathbf{z}(t), \boldsymbol{\tau})$. The log map has norm $\theta(t)$ and points along the target geodesic, so

$$\dot{\theta}(t) = -\frac{s(t)}{R(t)} \theta(t), \qquad \theta(t) = \theta(0) \frac{R(t)}{R(0)}. \tag{20}$$

With $s(t) = t$, this becomes $\theta(t) = \theta(0)(1 - t^2)$ and $\theta(1) = 0$. Exact arrival applies to the instance-only field. When $\beta > 0$, the relational term couples examples and may bend individual paths; the joint potential still decreases under the ideal continuous dynamics (Section 3.5).

3.4 Layer-Wise Flow Supervision

During training, the L Transformer layers provide states at $t_\ell = \ell/L$; no external solver evaluates Eq. 19. The fraction of remaining point-wise displacement assigned to transition ℓ is

$$\begin{aligned} \rho_\ell &= 1 - \exp\left(- \int_{t_\ell}^{t_{\ell+1}} \frac{s(u)}{R(u)}\, du \right) \\ &= 1 - \frac{R(t_{\ell+1})}{R(t_\ell)} \in (0, 1]. \end{aligned} \tag{21}$$

This integrated form remains well-defined on the final interval and gives $\rho_{L-1} = 1$ for any valid schedule.

At layer ℓ, we evaluate the field once and construct the next target as

$$\widetilde{\mathbf{Z}}^{(\ell+1)} = \operatorname{Exp}_{\mathbf{Z}^{(\ell)}} \left(\rho_\ell \mathbf{V}(\mathbf{Z}^{(\ell)}, \mathbf{T}, \mathbf{G}_T) \right). \tag{22}$$

For tangent vector $\mathbf{v}$, the spherical exponential map is

$$\operatorname{Exp}_{\mathbf{z}}(\mathbf{v}) = \cos(\|\mathbf{v}\|_2)\mathbf{z} + \sin(\|\mathbf{v}\|_2) \frac{\mathbf{v}}{\|\mathbf{v}\|_2}, \tag{23}$$

with its continuous extension at $\mathbf{v} = \mathbf{0}$. This update remains on the hypersphere for any local step size. For $\beta = 0$ and $\alpha = 1$, it is spherical interpolation toward the teacher by fraction ρ_ℓ. With the relational term, it is a one-step discretization of the coupled field, not an exact solution of the nonlinear trajectory.

The Transformer produces the actual state $\mathbf{Z}^{(\ell+1)}$. On the hypersphere, the actual movement made by the student from layer ℓ to $\ell + 1$ is the tangent vector

$$\mathbf{U}^{(\ell)} = \operatorname{Log}_{\mathbf{Z}^{(\ell)}} \left(\mathbf{Z}^{(\ell+1)} \right), \tag{24}$$

and the direction toward the teacher endpoint from the same state is

$$\mathbf{V}^{(\ell)} = \mathrm{Log}_{\mathbf{Z}^{(\ell)}}(\mathbf{T}), \qquad \mathbf{v}_i^{(\ell)} = \mathrm{Log}_{\mathbf{z}_i^{(\ell)}}(\boldsymbol{\tau}_i). \tag{25}$$

The endpoint-guided velocity loss asks each realized transition to point along this direction:

$$\mathcal{L}_{\text{evel}} = \frac{1}{L-1}\sum_{\ell=1}^{L-1}\frac{1}{B}\sum_{i=1}^{B}\Big[1-\cos\Big(\mathbf{u}_i^{(\ell)}, \mathbf{v}_i^{(\ell)}\Big)\Big], \tag{26}$$

where $\mathbf{u}_i^{(\ell)}$ and $\mathbf{v}_i^{(\ell)}$ are the rows of $\mathbf{U}^{(\ell)}$ and $\mathbf{V}^{(\ell)}$ and $\cos(\cdot,\cdot)$ is the cosine between tangent vectors. Both vectors live in the tangent space of $\mathbf{Z}^{(\ell)}$, so the comparison is well defined. The cosine is scale-free, so $\mathcal{L}_{\text{evel}}$ encourages each student layer to move in the tangent direction toward the final teacher embedding without prescribing an intermediate target state or a step magnitude. Every transition is covered, including the last one $\ell = L-1$ into the final layer: the endpoint objective below fixes *where* that layer must land, while $\mathcal{L}_{\text{evel}}$ asks that it be reached along the geodesic rather than by an arbitrary detour. In practice the direction is computed from a detached $\mathbf{Z}^{(\ell)}$, so the loss cannot be lowered by moving the source state to reorient the target.

3.5 Training Objective

The final layer must still satisfy the endpoint condition. We use

$$\mathcal{L}_{\text{end}} = \frac{1}{B}\sum_{i=1}^{B}\Big(1-(\mathbf{z}_i^{(L)})^\top \boldsymbol{\tau}_i\Big). \tag{27}$$

No earlier layer receives this direct target. The direction $\mathbf{v}_i^{(L-1)}$ points along the geodesic to $\boldsymbol{\tau}_i$, so $\mathcal{L}_{\text{end}}$ is the boundary condition of the same path: $\mathcal{L}_{\text{evel}}$ orients the final transition and $\mathcal{L}_{\text{end}}$ fixes where it must stop.

We also apply unsupervised contrastive learning at the output. For two dropout views $\mathbf{z}_i^{(L,a)}$ and $\mathbf{z}_i^{(L,b)}$,

$$\mathcal{L}_{\text{ctr}} = -\frac{1}{B}\sum_{i=1}^{B}\log\frac{\exp((\mathbf{z}_i^{(L,a)})^\top \mathbf{z}_i^{(L,b)}/\tau_c)}{\sum_{j=1}^{B}\exp((\mathbf{z}_i^{(L,a)})^\top \mathbf{z}_j^{(L,b)}/\tau_c)}. \tag{28}$$

Restricting this term to the output avoids a second layer-wise objective. The training loss is

$$\mathcal{L}_{\text{total}} = \lambda_{\text{end}}\mathcal{L}_{\text{end}} + \lambda_{\text{evel}}\mathcal{L}_{\text{evel}} + \lambda_{\text{ctr}}\mathcal{L}_{\text{ctr}}. \tag{29}$$

Here, λ_{end}, λ_{evel}, and λ_{ctr} weight endpoint, transition, and contrastive supervision. The potential $\mathcal{E}$ and its flow describe the ideal trajectory and drive the depth diagnostics; neither is an additional layer-wise loss. The term $\mathcal{L}_{\text{evel}}$ measures whether each Transformer layer moves along the geodesic toward the teacher endpoint.

Properties of the ideal flow. The construction gives three properties. First, the field is tangent to the product of hyperspheres. Consequently, $d\|\mathbf{z}(t)\|_2^2/dt = 2\mathbf{z}(t)^\top\dot{\mathbf{z}}(t) = 0$, so continuous trajectories preserve embedding norms. Second, for $t < 1$ the teacher-conditioned potential decreases monotonically:

$$\frac{d\mathcal{E}}{dt} = -\frac{B\,s(t)}{R(t)}\,\|\mathrm{grad}_{\mathbb{S}}\,\mathcal{E}\|_F^2 \le 0. \tag{30}$$

Third, Eq. 20 gives exact geodesic contraction for the instance-only field. These are properties of the ideal continuous flow; whether finite Transformer layers realize them is measured empirically. Full derivations are given in Appendix A.

Training and inference. Across L layers, the Gram term costs $\mathcal{O}(LB^2 d_S)$ training operations and $\mathcal{O}(B^2)$ temporary memory, independent of teacher size. Target construction and geometric supervision are removed at deployment, so inference uses the original student architecture and parameter count. Implementation details are provided in Appendix C.

4 Experiments

Table 1: Main downstream results across three teacher–student configurations. Classification datasets are evaluated with F1, pair-classification datasets with average precision (AP), and semantic textual similarity datasets with Spearman correlation. Higher is better. Within each configuration, the best result among distillation methods is shown in bold and the second-best result is underlined; Teacher and Student base are excluded from this ranking.

	Classification (F1)			Pair Classification (AP)			STS (Spearman)			
Method	Banking77	Tweet	Emotion	MRPC	SciTail	WiC	SICK	STS12	STS-B	Avg.
(a) Qwen3-Embedding-4B → BERT-base (109M)										
Teacher	93.63	72.22	68.94	84.79	91.77	66.80	83.43	82.55	86.87	81.22
Student base	84.86	47.79	68.02	77.36	67.74	59.22	42.43	21.54	20.30	54.36
Stella	91.08	73.40	64.40	<u>86.23</u>	83.22	<u>68.98</u>	78.38	71.31	79.16	77.35
CDM	89.71	70.76	52.67	84.03	71.47	**69.83**	71.36	68.87	74.68	72.60
DSKD	88.89	69.31	53.39	**86.35**	67.73	68.89	70.58	67.99	74.29	71.94
EMO	85.92	69.39	55.41	85.79	64.34	67.73	63.79	64.41	70.14	69.60
TALAS	**93.34**	<u>74.45</u>	**72.70**	86.16	**87.78**	68.11	<u>80.70</u>	**77.34**	**83.57**	**80.46**
GeoODE-KD	<u>93.22</u>	**74.53**	<u>72.39</u>	85.92	<u>86.13</u>	67.01	**81.17**	<u>75.58</u>	<u>81.27</u>	<u>79.69</u>
(b) BGE-M3 → MiniLMv2-H768 (66M)										
Teacher	93.52	73.85	68.56	85.81	91.87	61.47	79.18	78.73	84.87	79.76
Student base	81.84	47.70	71.67	79.67	65.06	60.89	49.46	26.76	24.12	56.35
Stella	86.62	71.08	58.57	83.63	75.85	69.38	70.87	63.08	67.88	71.89
CDM	87.96	71.31	58.31	83.83	75.43	<u>69.58</u>	67.37	62.12	67.09	71.44
DSKD	88.45	70.93	58.96	83.25	75.96	69.53	66.19	61.50	65.53	71.14
EMO	87.61	71.19	59.27	82.90	75.03	**69.69**	65.93	57.46	62.75	70.20
TALAS	<u>90.14</u>	<u>74.91</u>	<u>63.50</u>	<u>86.76</u>	<u>81.27</u>	66.53	<u>76.31</u>	<u>67.66</u>	<u>78.18</u>	<u>76.14</u>
GeoODE-KD	**92.37**	**76.70**	**66.93**	**86.96**	**86.40**	63.53	**78.84**	**72.56**	**80.32**	**78.29**
(c) Qwen3-Embedding-0.6B → MiniLMv2-H384 (22M)										
Teacher	93.21	71.33	66.53	83.37	90.00	66.85	80.64	77.11	84.60	79.29
Student base	68.29	41.07	69.90	78.39	64.56	59.58	47.60	21.95	22.08	52.60
Stella	84.36	68.81	54.01	81.61	72.69	<u>68.53</u>	65.30	61.02	64.70	69.00
CDM	85.21	69.36	52.31	81.65	73.12	68.52	65.20	61.91	65.23	69.17
DSKD	85.28	69.34	53.41	81.40	72.77	**68.56**	64.88	61.42	64.85	69.10
EMO	84.40	69.20	56.13	81.68	72.21	68.22	63.82	57.88	63.34	68.54
TALAS	<u>86.67</u>	<u>72.41</u>	<u>59.93</u>	<u>84.75</u>	<u>80.47</u>	66.72	<u>71.59</u>	<u>69.80</u>	<u>74.94</u>	<u>74.14</u>
GeoODE-KD	**91.44**	**74.24**	**66.59**	**85.28**	**82.47**	67.03	**77.19**	**72.33**	**77.71**	**77.14**

5 Discussion and Limitations

6 Conclusion

AI Use Statement

Ethics Statement

Reproducibility Statement

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A Mathematical Derivations and Proofs

A.1 Riemannian Geometry and Vector Field Derivations

The batch state lies on the product manifold

$$\mathcal{M} = \left(\mathbb{S}^{d_S-1}\right)^B, \tag{31}$$

equipped with the Frobenius inner product. Its tangent space at $\mathbf{Z}$ is

$$T_{\mathbf{Z}}\mathcal{M} = \left\{\mathbf{U} \in \mathbb{R}^{B\times d_S} : \mathbf{z}_i^\top \mathbf{u}_i = 0 \text{ for every } i\right\}. \tag{32}$$

The row-wise projection in Eq. 15 follows directly:

$$\Pi_{\mathbf{Z}}(\mathbf{U})_i = \mathbf{u}_i - (\mathbf{z}_i^\top \mathbf{u}_i)\mathbf{z}_i. \tag{33}$$

Point-wise semantic gradient. For one normalized pair $(\mathbf{z}, \boldsymbol{\tau})$, let $c = \mathbf{z}^\top \boldsymbol{\tau}$ and $\theta = \arccos(c) \in [0, \pi)$. We exclude the antipodal cut locus $\theta = \pi$, where the shortest geodesic is not unique. The ambient derivative of the squared geodesic potential is

$$\nabla_{\mathbf{z}} \frac{1}{2}\theta^2 = -\frac{\theta}{\sin\theta}\boldsymbol{\tau}. \tag{34}$$

Projecting this derivative onto the tangent space gives

$$\begin{aligned} \operatorname{grad}_{\mathbb{S}} \frac{1}{2}\theta^2 &= -\frac{\theta}{\sin\theta}(\boldsymbol{\tau} - c\mathbf{z}) \\ &= -\operatorname{Log}_{\mathbf{z}}(\boldsymbol{\tau}). \end{aligned} \tag{35}$$

The continuous extension at $\theta = 0$ is zero. Applying Eq. 35 to the batch mean in Eq. 12 yields

$$\operatorname{grad}_{\mathbb{S}} \mathcal{E}_{\text{sem}}(\mathbf{Z}, \mathbf{T}) = -\frac{1}{B}\operatorname{Log}_{\mathbf{Z}}(\mathbf{T}). \tag{36}$$

The norm of each logarithmic-map row equals its geodesic distance because

$$\|\boldsymbol{\tau} - c\mathbf{z}\|_2 = \sin\theta, \qquad \|\operatorname{Log}_{\mathbf{z}}(\boldsymbol{\tau})\|_2 = \theta. \tag{37}$$

Relational gradient. Let

$$\mathbf{A} = \mathbf{Z}\mathbf{Z}^\top - \mathbf{G}_T. \tag{38}$$

Both Gram matrices are symmetric, so $\mathbf{A} = \mathbf{A}^\top$. For a perturbation $\mathbf{H}$, the differential of Eq. 13 is

$$\begin{aligned} \mathrm{D}\mathcal{E}_{\text{geo}}[\mathbf{H}] &= \frac{2}{B^2}\left\langle \mathbf{A}, \mathbf{H}\mathbf{Z}^\top + \mathbf{Z}\mathbf{H}^\top \right\rangle_F \\ &= \frac{4}{B^2}\left\langle \mathbf{A}\mathbf{Z}, \mathbf{H} \right\rangle_F. \end{aligned} \tag{39}$$

Hence the ambient and Riemannian gradients are

$$\begin{aligned} \nabla_{\mathbf{Z}}\mathcal{E}_{\text{geo}} &= \frac{4}{B^2}\mathbf{A}\mathbf{Z}, \\ \operatorname{grad}_{\mathbb{S}} \mathcal{E}_{\text{geo}} &= \frac{4}{B^2}\Pi_{\mathbf{Z}}(\mathbf{A}\mathbf{Z}). \end{aligned} \tag{40}$$

Combining Eqs. 36 and 40 gives

$$\operatorname{grad}_{\mathbb{S}} \mathcal{E} = -\frac{\alpha}{B}\operatorname{Log}_{\mathbf{Z}}(\mathbf{T}) + \frac{4\beta}{B^2}\Pi_{\mathbf{Z}}\left[(\mathbf{Z}\mathbf{Z}^\top - \mathbf{G}_T)\mathbf{Z}\right]. \tag{41}$$

Multiplying by $-B$ recovers the vector direction in Eq. 17. This scaling removes the $1/B$ from the batch mean without changing the descent direction.

A.2 Finite-Horizon Flow Derivation

Consider first the teacher-conditioned gradient flow in an auxiliary time τ,

$$\frac{d\mathbf{Z}}{d\tau} = \mathbf{V}(\mathbf{Z}, \mathbf{T}, \mathbf{G}_T) = -B \operatorname{grad}_{\mathbb{S}} \mathcal{E}. \tag{42}$$

This flow generally reaches a stationary point only as $\tau \to \infty$. To fit that horizon into student depth $t \in [0, 1]$, define

$$\tau(t) = \int_0^t \frac{s(u)}{R(u)} \, du, \qquad R(t) = \int_t^1 s(u) \, du. \tag{43}$$

Since $\dot{R}(t) = -s(t)$,

$$\frac{d}{dt} \log R(t) = -\frac{s(t)}{R(t)}, \tag{44}$$

and therefore

$$\tau(t) = \log \frac{R(0)}{R(t)}. \tag{45}$$

For any schedule with $R(t) > 0$ for $t < 1$ and $R(1) = 0$, $\tau(t) \to \infty$ as $t \to 1$. Applying the chain rule to Eq. 42 gives

$$\frac{d\mathbf{Z}}{dt} = \frac{d\tau}{dt} \frac{d\mathbf{Z}}{d\tau} = \frac{s(t)}{R(t)} \mathbf{V}(\mathbf{Z}, \mathbf{T}, \mathbf{G}_T), \tag{46}$$

which is Eq. 19.

Schedules. For $s(t) = t^p$ with $p > -1$,

$$R(t) = \frac{1 - t^{p+1}}{p + 1}. \tag{47}$$

The constant schedule is $p = 0$, while the default linear schedule is $p = 1$ and gives $R(t) = (1 - t^2)/2$.

Discrete step fraction. The warped time elapsed between two student depths is

$$\begin{aligned} \Delta\tau_\ell &= \int_{t_\ell}^{t_{\ell+1}} \frac{s(u)}{R(u)} \, du \\ &= \log \frac{R(t_\ell)}{R(t_{\ell+1})}. \end{aligned} \tag{48}$$

Under the instance-only auxiliary flow, the distance remaining after this interval is multiplied by $\exp(-\Delta\tau_\ell)$. The fraction covered is therefore

$$\begin{aligned} \rho_\ell &= 1 - \exp(-\Delta\tau_\ell) \\ &= 1 - \frac{R(t_{\ell+1})}{R(t_\ell)}, \end{aligned} \tag{49}$$

which recovers Eq. 21. For the default schedule and $t_\ell = \ell/L$,

$$\rho_\ell = \frac{2\ell + 1}{L^2 - \ell^2}, \qquad \rho_{L-1} = 1. \tag{50}$$

For the point-wise field, this fraction reproduces the exact geodesic progress over the interval. For the coupled semantic–relational field, GeoODE-KD uses the same ρ_ℓ as a local exponential-map step; it is a discretization, not an exact integration of the changing nonlinear field.

The exponential map also preserves the discrete target norm. If $\mathbf{v} \in T_{\mathbf{z}}\mathbb{S}^{d_S - 1}$, then $\mathbf{z}^\top \mathbf{v} = 0$, and Eq. 23 gives

$$\|\mathrm{Exp}_{\mathbf{z}}(\mathbf{v})\|_2^2 = \cos^2 \|\mathbf{v}\|_2 + \sin^2 \|\mathbf{v}\|_2 = 1. \tag{51}$$

A.3 Proofs of the Flow Properties

The following statements assume $s(t) \geq 0$, $R(t) > 0$ for $t < 1$, and that no student–teacher pair lies at the antipodal cut locus. The logarithmic map is defined by continuous extension when a student state already equals its target.

Proposition 1 (norm preservation). Every row of a solution to Eq. 19 remains on the unit hypersphere.

Proof. Both terms in Eq. 17 are tangent: the log map lies in $T_{\mathbf{z}_i}\mathbb{S}^{d_S-1}$, and the relational term is explicitly projected by $\Pi_{\mathbf{Z}}$. Thus $\mathbf{z}_i^\top \dot{\mathbf{z}}_i = 0$ and

$$\frac{d}{dt}\|\mathbf{z}_i(t)\|_2^2 = 2\mathbf{z}_i(t)^\top \dot{\mathbf{z}}_i(t) = 0. \tag{52}$$

Since $\|\mathbf{z}_i(0)\|_2 = 1$, its norm remains one. $\square$

Proposition 2 (energy descent). Along the ideal continuous flow, the teacher-conditioned potential is non-increasing for every $t < 1$.

Proof. By Eq. 41, $\mathbf{V} = -B\,\mathrm{grad}_{\mathbb{S}}\,\mathcal{E}$. The Riemannian chain rule gives

$$\begin{aligned}\frac{d\mathcal{E}}{dt} &= \left\langle \mathrm{grad}_{\mathbb{S}}\,\mathcal{E}, \dot{\mathbf{Z}} \right\rangle_F \\ &= -\frac{B\,s(t)}{R(t)}\,\|\mathrm{grad}_{\mathbb{S}}\,\mathcal{E}\|_F^2 \leq 0.\end{aligned} \tag{53}$$

This proves Eq. 30. $\square$

Corollary 1 (point-wise geodesic contraction). For $\alpha = 1$ and $\beta = 0$, the geodesic distance $\theta(t) = d_g(\mathbf{z}(t), \boldsymbol{\tau})$ satisfies

$$\theta(t) = \theta(0)\frac{R(t)}{R(0)}, \qquad \lim_{t\to 1}\theta(t) = 0. \tag{54}$$

Proof. For one pair, define $e(t) = \frac{1}{2}\theta(t)^2$. From Eq. 35 and the point-wise form of Eq. 19,

$$\begin{aligned}\dot{e}(t) &= \left\langle -\mathrm{Log}_{\mathbf{z}}(\boldsymbol{\tau}), \frac{s(t)}{R(t)}\,\mathrm{Log}_{\mathbf{z}}(\boldsymbol{\tau}) \right\rangle \\ &= -\frac{s(t)}{R(t)}\theta(t)^2,\end{aligned} \tag{55}$$

where Eq. 37 is used in the last line. For $\theta(t) > 0$, $\dot{e} = \theta\dot{\theta}$, so

$$\frac{d}{dt}\log\theta(t) = -\frac{s(t)}{R(t)} = \frac{d}{dt}\log R(t). \tag{56}$$

Integrating from 0 to t proves Eq. 54. If θ reaches zero earlier, the log-map vector is zero and the state remains at the target. $\square$

If the observed student trajectory begins at the first Transformer layer $t_1 = 1/L$ rather than at $t = 0$, the same argument starts from t_1 and gives

$$\theta(t) = \theta(t_1)\frac{R(t)}{R(t_1)}, \qquad t \geq t_1. \tag{57}$$

Since $\theta(t) \in [0, \pi)$ and $\dot{\theta}(t) \leq 0$, teacher cosine is also non-decreasing:

$$\frac{d}{dt}\cos\theta(t) = \frac{s(t)}{R(t)}\theta(t)\sin\theta(t) \geq 0. \tag{58}$$

Corollary 2 (exact sampled profile). For $0 \leq a \leq 1$, the curve

$$\gamma(a) = \mathrm{Exp}_{\mathbf{z}}(a \,\mathrm{Log}_{\mathbf{z}}(\boldsymbol{\tau})) \tag{59}$$

is the shortest geodesic from $\mathbf{z}$ to $\boldsymbol{\tau}$ and satisfies

$$d_g(\gamma(a), \boldsymbol{\tau}) = (1 - a) d_g(\mathbf{z}, \boldsymbol{\tau}). \tag{60}$$

Setting $a = \rho_\ell$ in the instance-only target gives

$$\theta_{\ell+1} = (1 - \rho_\ell)\theta_\ell = \frac{R(t_{\ell+1})}{R(t_\ell)}\theta_\ell. \tag{61}$$

The factors telescope, so the predicted states reproduce Eq. 54 at every sampled depth and the last target equals $\boldsymbol{\tau}$ because $\rho_{L-1} = 1$.

Scope of the guarantees. Norm preservation and energy descent describe the ideal continuous field. Exact arrival and the sampled profile require the instance-only setting $\alpha = 1, \beta = 0$. With $\beta > 0$, the native teacher Gram target can compete with the projected point-wise target, so individual distances need not decrease at every depth. Finally, the realized Transformer trajectory satisfies the sampled profile only to the extent that it minimizes $\mathcal{L}_{\mathrm{evel}}$; the propositions do not assume zero discretization or optimization error.

B Target Construction and Gauge Alignment

B.1 PCA Initialization

B.2 Gram-Consistent Target Fitting

B.3 Orthogonal Gauge Alignment

C Training Algorithm and Implementation Details

D Datasets, Baselines, and Evaluation Protocol

E Additional Results and Depth Diagnostics

F Extended Ablations and Sensitivity Analysis